Anupoju Venkata Naga Sai Swathi Kiran

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.64
NARAYANA JUNIOR COLLEGE, VIJAYAWADA	Class XII (ANDHRA PRADESH BOARD OF INTERMEDIATE EDUCATION)	2013	92.10%
S R NATIONAL HIGH SCHOOL, WARANGAL	Class X (ANDHRA PRADESH BOARD OF SECONDARY EDUCATION)	2011	90.00%

INDUSTRY-COLLABORATIVE ENGINEERING PROJECTS

Bird-Strike Impact Testing

- Supported high-speed **bird-strike testing** using bird and gelatin projectiles at velocities up to **200 m/s** under representative flight conditions.
- Generated and analyzed **impact-response** data to support the design and validation of **bird-strike-resistant aerospace structures**.
- Supported test execution, data analysis, and interpretation of **projectile-structure interaction** for aerospace safety applications.

Dynamic Material Characterization

- Conducted **high-strain-rate testing of E250, SS304, and tungsten alloys** using dynamic mechanical characterization techniques, evaluating **strain-rate-dependent deformation, strength, and failure behavior**. For E250, performed SHPB testing up to **8000 /s** and developed **Johnson-Cook strength and damage parameters** for numerical simulations
- Conducted experimental investigation of **porous gypsum under compression and indirect tension** across a wide range of strain rates using **UTM and SHPB**, evaluating the effects of porosity on **strength, strain-rate sensitivity, Dynamic Increase Factor (DIF), and Young's modulus**.
- Investigated the **high-strain-rate tensile behavior and failure of adhesive joints** using a **Split Hopkinson Tensile Bar (SHTB)**, characterizing joint response and failure mechanisms under dynamic loading

Tyre Impact Testing & Dynamic Response Characterization:

- Conducted **dynamic impact testing of MRF tyres** using a drop-impact machine across **7-11.5 m/s impact velocities**, evaluating force, displacement, and velocity responses under different striker configurations.
- Integrated **high-speed camera/DIC measurements with machine data** to characterize tyre deformation and impact response, assess repeatability, and identify failure conditions; established a failure threshold at **8 m/s after two successive impacts** under 45 psi tyre pressure and 220 kg preload.

Low-Velocity Impact Testing

- Experimental investigation of **Carbon fiber, Kevlar fiber, and Aluminium foam sandwich composite** panels using an instrumented drop-weight tower.
- Evaluated the **energy absorption**, peak load impact, and damage response of sandwich panels for **impact-resistant structural applications**.
- Delivered experimental results and engineering insights for a government-sponsored research work on **light-weight protective structures**.

3D-Printed Lattice structures for heat transfer applications

- Designed, fabricated through **Laser Powder Bed Fusion**, and experimentally investigated the heat transfer efficiency of **lattice structures**.

- Developed a custom **forced-convection test rig** to evaluate the **lattice thermal performance** for various lattice geometries and air flow conditions.

ACADEMIC PROJECTS

PhD (Transient Pressure Loading Systems and Rupture Disc Design)

IIT Hyderabad | 2020 - Present

- Designed and developed a custom experimental setup for **transient pressure loading** of clamped circular plates and metallic structures.
- Designed novel rupture-disc configurations and evaluated structural performance through simulation-driven design and experimental testing.
- Developed and validated **ABAQUS/Explicit Coupled Eulerian-Lagrangian (CEL)** models for **large deformation**, **dynamic loading**, and failure prediction.
- Developed **Fortran** user subroutines (**VDLOAD**) to implement experimentally representative transient pressure loading.
- Developed and experimentally validated **Cohesive Zone Models (CZM)** for adhesively bonded structural components.
- Performed **nonlinear structural analysis**, dynamic response simulations, **failure investigations**, and **simulation-experiment correlation** to support design decisions.
- Applied **DFMEA** principles to identify failure modes and evaluate rupture-disc design configurations.
- Developed CFRP composite discs to use them as burst discs and evaluate the performance of CFRP composites in the brust disc industry.
- Working on **AI/ML models** to predict the burst pressure of the pre-cut disc for given variables.

M.Tech (Tribological Study of Al-based Metal Matrix Composites)

NIT Warangal | 2018 - 2020

- **Developed Al-CrB2/TiB2** metal matrix **composites** through pressure-less sintering and investigated **reciprocative sliding wear** response of the composites.
- Performed **XRD, SEM, EBSD** characterization and **tribological testing** of the developed composites.
- Investigated **wear mechanisms** and correlated material properties with **tribological performance** using optical and SEM image analysis.

TECHNICAL EXPERTISE

Mechanical Design, Composites, Impact Mechanics, Experimental Mechanics, 3D- printing, Heat Transfer, Failure Analysis, Linear/Non-Linear FEA

TECHNICAL SKILLS

CAE/FEA: ABAQUS, LS-DYNA, ANSYS, EXPLICIT ANALYSIS, CEL, CZM, FAILURE ANALYSIS, COMPOSITES, Non-linear FEA

Designing and Modeling: SOLIDEDGE, AUTOCAD

Programming: MATLAB, Python, C

Engineering Methods: DFMEA, DFMA, Structural Mechanics, Impact Mechanics, Heat Transfer, Experimental Mechanics

Manufacturing: Metal additive manufacturing/ 3D Printing

RESEARCH CONTRIBUTION

- "Development of an experimental setup for testing rupture discs" at the 13th International Symposium on Plasticity and Impact Mechanics (IMPLAST) at IIT Madras in 2022
- "Numerical investigation of the effect of yield strength on the behavior of a clamped circular plate under transient pressure loading" at the 21st International Conference on Experimental Mechanics (ICEM21), conducted at Bologna, Italy, in 2025
- "Analysis of Pre-cut Burst discs" at IMPLAST conference conducted at IIT Roorkee in 2025

LEADERSHIP AND RESPONSIBILITIES

Elected **Department Representative (B.Tech)** - Represented 120+ students; coordinated with faculty on academic and administrative matters.

Class Representative (M.Tech) - Led class communications, organized seminars, and supported academic planning

Team Lead - Brakes System, **E-BAJA** competition (**SAE India & Mahindra & Mahindra**)

EXTRACURRICULAR INTERESTS & PERSONAL PURSUITS

Sports & Teamwork: Enthusiastic volleyball and cricket player, passionate about team sports, strategic gameplay, and competitive spirit.

Television & Pop Culture: A passionate follower of iconic TV series, including *FRIENDS, Breaking Bad, Game of Thrones,* and *The Big Bang Theory*.

Marvel Cinematic Universe: An enthusiastic Marvel fan with a strong interest in its interconnected storytelling, characters, and cinematic universe.